

Product Requirements Document (PRD)

Product Name

Magic OTP Asset Access Platform – Version 2.0

Objective

Build a secure, interactive, and intelligent platform where employees can scan QR codes on their assigned laptops/accessories and access detailed asset information using time-limited OTPs—without traditional password login. Version 2.0 will integrate AI-driven security and usability enhancements while retaining core Magic OTP functionality.

1. Target Users

- **Employees:** View assigned asset and accessory details by scanning QR.
 - **Admins/IT Team:** Manage assets, generate QR codes, monitor access logs, and receive anomaly alerts.
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2. Updated Key Features

A. Core Workflow (Magic OTP)

1. Employee scans QR code on a device.
2. App sends QR slug to backend (`/otp/request`).
3. OTP is generated and sent via SMTP to the employee's registered e-mail.
4. Employee enters OTP to `/otp/verify` .
5. Upon verification, a short-lived JWT is issued (15 minutes).
6. Device details and accessory list are displayed.

B. AI-Enhanced Features

1. **Anomaly Detection:** Isolation Forest model flags unusual access (e.g., midnight scan, new IP).
 2. **OTP Abuse Detection:** Monitors rate and frequency of OTP requests using LSTM or clustering.
 3. **Voice-based OTP Input (Mobile):** React Native integration for voice entry of OTP.
 4. **Admin Chat Assistant (Optional):** LangChain-powered query assistant over audit logs.
 5. **Email Priority Score:** AI model prioritizes email retry logic based on user's email open behavior.
 6. **Load Prediction:** Forecast spikes in asset scans to pre-scale Redis and Celery worker pools.
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3. Tech Stack

Layer	Tool/Service
Backend	FastAPI, Celery, Python 3.12
Frontend	React 18, TailwindCSS, React Native
Auth	Magic OTP + HS256 JWT
Email	SMTP (Gmail/O365)
Cache	Redis 7
DB	PostgreSQL 16
AI/ML	scikit-learn, Prophet, LangChain, OpenAI APIs
Vector DB	FAISS or PGVector (for chatbot)
Infra	Docker, Kubernetes, Helm, Prometheus, Grafana

4. Workflow Diagram

```
[Employee Scan] → /otp/request
├─ Redis Rate Limit Bucket
├─ Anomaly Detection Check
└─ Send OTP (SMTP)

[Employee Enters OTP] → /otp/verify
├─ Validate OTP (Redis)
├─ OTP Abuse Detection Model
└─ Issue JWT if verified

[JWT Authenticated Access] → /asset/:id
└─ Return asset + accessory data
```

5. API Endpoints (Extended)

Method	Path	Auth	Description
POST	<code>/otp/request</code>	None	Send OTP for scanned QR slug
POST	<code>/otp/verify</code>	None	Verify OTP, issue JWT

Method	Path	Auth	Description
GET	<code>/asset/:id</code>	JWT	Fetch asset + accessory details
POST	<code>/asset</code>	Admin JWT	Add asset + accessories
PATCH	<code>/qr/:slug/revoke</code>	Admin JWT	Revoke QR permanently
GET	<code>/admin/logs</code>	Admin JWT	Filterable audit log retrieval
POST	<code>/admin/query-bot</code>	Admin JWT	Natural language query (chatbot)

6. Data Requirements

- Asset details (RAM, storage, brand, model, etc.)
- User (employee) email/phone mapping
- OTP session history and verification attempts
- Audit trail (who accessed what, when, and from where)
- Logs for AI training (timestamps, IPs, device IDs, etc.)

7. Security Controls

Control	Value
OTP TTL	5 minutes
OTP Attempts	Max 3 per session
JWT Expiry	15 minutes (no refresh)
Redis Rate Limits	5 scans/min/user
IP-level Flood Protection	100 req / 15 min / IP
HTTPS Enforcement	Required
QR Revocation	Supported via admin panel

8. Testing Strategy

- Unit tests (pytest + coverage)
- E2E Tests (Cypress & Detox)
- Load Test (Locust)
- AI model eval: Precision, recall, F1 for anomaly flags
- Security scan: Bandit, OWASP ZAP

9. Milestones

Week	Task
1	Setup repo, Docker, DB migrations
2	Asset + QR APIs, frontend scan flow
3	OTP logic + Redis + Email worker
4	Rate limiting, audit logs, React OTP UI
5	AI: Anomaly detection + voice OTP
6	Admin dashboard + chatbot (optional)
7	Final testing, deploy to prod

10. Future Enhancements

- SMS fallback for OTP
- Asset check-in/check-out flow
- Auto-revocation after inactivity
- Real-time dashboard for access patterns
- Multilingual support for frontend

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